METHODOLOGY – COMBINED VALUE AND MOMENTUM MODEL

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1. Overview of the Methodological Framework

This study constructs a systematic quantitative equity model for the Vietnamese stock market by integrating two well-established return premia: the value premium and the momentum premium. Drawing primarily from Asness, Moskowitz, and Pedersen (2013), who document the pervasiveness of value and momentum return premia across eight global asset classes, this methodology adapts their cross-sectional framework to the specific institutional, regulatory, and market microstructure characteristics of Vietnam.

The model encompasses three distinct portfolio strategies: (i) a value-only model rebalanced quarterly, (ii) a momentum-only model rebalanced weekly, and (iii) a combined model integrating both factors at weekly frequency. Each strategy constructs three portfolios through a within-sector tertile sort, labelled P1 (top third), P2 (middle third), and P3 (bottom third), following the spirit of Fama and French (1993) in their construction of size and book-to-market portfolios. Portfolio returns are computed using market capitalisation weighting to reflect the investable universe of a large institutional investor.

Several adaptations are made relative to the original framework of Asness et al. (2013) to accommodate Vietnamese market conditions. First, short selling is excluded from all portfolio constructions given the highly restricted and practically infeasible short-selling environment in Vietnam, rendering the model long-only throughout. Second, sector-specific value measures are employed, using the price-to-book (P/B) ratio for financial and banking stocks and the price-to-earnings (P/E) ratio for all other sectors, reflecting the differences in balance sheet structures across industries documented by Lakonishok, Shleifer, and Vishny (1994). Third, a 45-day publication lag is applied to all quarterly fundamental data to eliminate look-ahead bias, consistent with the approach of Fama and French (1992).

2. Data

2.1 Price Data

Daily adjusted closing prices and trading volumes are obtained for 301 stocks included in VNALL-INDEX, covering the period from January 2015 to May 2026. All price series are adjusted for corporate actions including stock splits, reverse splits, and rights issues.

The cleaned daily price matrix is converted into wide format with dates as rows and tickers as columns. Weekly prices are constructed using the last available Thursday closing price. This weekly Thursday series is used for the momentum signal and weekly holding-period returns. The script also creates monthly price and return matrices for reference, but the final value-model return computation is based on direct returns between quarterly rebalance dates rather than monthly aggregation.

2.2 Fundamental Data

Quarterly financial data including the price-to-earnings (P/E) ratio and price-to-book (P/B) ratio are obtained from FiinProX, covering the period from Q1 2021 to Q1 2026 for all 301 stocks in the VNALL-INDEX. The data are provided in wide format with quarterly observations as columns, and are reshaped into long format during processing. In line with the conservative data availability practices recommended by Fama and French (1992), a publication lag of 45 days is applied to all quarterly observations, meaning Q1 data (quarter ending March 31) is made available to the model from May 15 at the earliest. This procedure ensures that the model only uses fundamental data that was genuinely available to investors at the time of each portfolio formation date, thereby eliminating any look-ahead bias.

Any quarterly observation in which both the P/E and P/B values are missing is excluded from the dataset. Negative P/E values are excluded from the value signal computation, as they reflect loss-making periods in which the earnings yield is economically meaningless (Lakonishok et al., 1994). Additionally, P/E ratios exceeding 200 times are excluded as they typically indicate transitorily depressed earnings rather than genuine cheapness (Asness et al., 2013).

2.3 Market Capitalisation Data

Daily market capitalisation data are obtained from FiinProX for all stocks in the universe, covering the same sample period. Market capitalisation is defined as the product of the closing share price and the number of shares outstanding on the corresponding date. For each weekly portfolio formation date, the most recently available market capitalisation observation on or before that date is used, applying a forward-fill procedure analogous to that used for fundamental data. This approach prevents the use of future information in the construction of portfolio weights.

3. Signal Construction

3.1 Value Signal

The value signal is sector-specific. For stocks classified as Financials or Banks, the signal is book yield, defined as the inverse of the price-to-book ratio. For all other sectors, the signal is earnings yield, defined as the inverse of the price-to-earnings ratio. This distinction reflects the different economic interpretation of book value in financial institutions compared with non-financial firms, consistent with the motivation in Fama and French (1992, 1993) and Lakonishok, Shleifer and Vishny (1994).

$$x_{i,t}^{V} = \frac{1}{PB_{i,t}}, \quad \text{if } s(i) \in \{\text{Financials,Banks}\} \tag{1}$$

$$x_{i,t}^{V} = \frac{1}{PE_{i,t}}, \quad \text{if } s(i) \notin \{\text{Financials,Banks}\} \tag{2}$$

where $x_{i,t}^{V}$ denotes the value signal of stock i at date t, $PB_{i,t}$ is the price-to-book ratio, $PE_{i,t}$ is the price-to-earnings ratio, and $s(i)$ denotes the sector classification of stock i. For stocks classified as Financials or Banks, the value signal is book yield, $1/PB_{i,t}$. For all other sectors, the value signal is earnings yield, $1/PE_{i,t}$. A higher value signal therefore indicates a cheaper stock relative to its valuation anchor.

A higher value signal indicates a cheaper stock. At each quarterly value formation date, the model selects the most recently available valid value signal for each ticker whose safe date is less than or equal to the formation date. The value-only model then ranks stocks within sectors using this signal, with higher values assigned to P1.

3.2 Momentum Signal

The momentum signal is implemented as a short-horizon weekly cross-sectional momentum measure. The weekly price series is anchored to Thursday closing prices. At each weekly formation date t, the momentum signal for stock i is defined as the simple return from the previous Thursday close to the current Thursday close:

$$x_{i,t}^{M} = \frac{P_{i,t}^{Thu}}{P_{i,t-1}^{Thu}} - 1 \tag{3}$$

where $x_{i,t}^{M}$ denotes the momentum signal of stock i at weekly formation date t, $P_{i,t}^{Thu}$ is the Thursday closing price of stock i in the current week, and $P_{i,t-1}^{Thu}$ is the Thursday closing price of the same stock in the previous week.

The signal is observed after the current Thursday close, and the rebalancing action is assumed to take place on Friday of the same week. This timing convention ensures that the portfolio is formed only after the full Thursday-to-Thursday return is observable. The use of momentum follows the cross-sectional momentum literature initiated by Jegadeesh and Titman (1993) and extended in multi-asset settings by Asness, Moskowitz and Pedersen (2013), but the horizon is deliberately adapted to the final weekly implementation.

3.3 Weekly Value Signal for the Combined Model

The combined model requires both a value signal and a momentum signal at weekly frequency. Because the value signal is based on quarterly fundamentals, our methodology forward-fills the most recently valid quarterly value signal to each weekly date. A stock is eligible for the combined model only when it has a valid weekly momentum signal, a valid forward-filled value signal, and valid market capitalisation at the weekly formation date.

4. Portfolio Construction

4.1 Within-Sector Tertile Sort

All models use within-sector ranking to reduce unintended sector bets and to avoid ranking stocks across sectors using signals with different empirical distributions. At each formation date, stocks are grouped by ICB Level 1 sector and ranked by the relevant signal. The best floor (n/3) stocks in each sector are assigned to P1, the worst floor(n/3) stocks are assigned to P3, and the remaining stocks are assigned to P2. This approach follows the spirit of sector-aware factor construction in the empirical asset-pricing literature, including Fama and French (1993), Asness, Moskowitz and Pedersen (2013), and Israel and Moskowitz (2012).

The minimum sector size is set to six stocks. If a sector has fewer than six valid stocks at a given formation date, all stocks in that sector are assigned to P2. This avoids constructing top and bottom tertiles from very small sector samples. Because the middle portfolio absorbs both remainder stocks and small-sector stocks, P2 is typically larger than P1 and P3.

4.2 Combined Value-Momentum Score

For the combined model, the value and momentum signals are first standardised within each sector using z-scores. This standardisation is necessary because the value and momentum signals are measured on different scales. It also ensures that the combined score is not mechanically dominated by the signal with the larger numerical dispersion.

Let $x_{i,t}^{V}$ denote the value signal of stock i at formation date t, and let $x_{i,t}^{M}$ denote the momentum signal. Let $s(i)$ denote the sector classification of stock i. For each signal $f \in \{V, M\}$, the within-sector z-score is computed as:

$$z_{i,t}^{f} = \frac{x_{i,t}^{f} - \mu_{s(i),t}^{f}}{\sigma_{s(i),t}^{f}} \tag{4}$$

where $\mu^{f}_{s(i),t}$ and $\sigma^{f}_{s(i),t}$ are the cross-sectional mean and standard deviation of signal f among stocks in the same sector as stock i at formation date t. If the sector-level standard deviation is zero or missing, the corresponding z-score is set to zero. Each z-score is then clipped to the interval $[-3,3]$ to reduce the influence of extreme observations.

The combined value-momentum score is computed as an equal-weighted average of the value and momentum z-scores:

$$C_{i,t} = 0.5z^{V}_{i,t} + 0.5z^{M}_{i,t} \tag{5}$$

where $C_{i,t}$ denotes the combined score of stock i at formation date t. Stocks are then ranked within each sector by $C_{i,t}$ and assigned to P1, P2, or P3 using the same within-sector tertile procedure described above. A higher combined score indicates a stock that is simultaneously cheaper on the value signal and stronger on the weekly momentum signal.

The equal weighting of the two standardised signals follows the empirical motivation in Asness, Moskowitz, and Pedersen (2013), who document that value and momentum contain complementary information across asset classes. In the Vietnamese market implementation, this structure allows the model to combine slow-moving valuation information with short-horizon price-trend information while maintaining sector comparability.

4.3 Sector-Replicated Market-Capitalisation Weighting

The final market-capitalisation-weighted portfolios do not use simple portfolio-level market-capitalisation weights. Instead, they apply a sector-replicated market-capitalisation weighting scheme. This approach first estimates each sector’s weight in the full eligible investment universe, then reallocates capital across the sectors represented in the target portfolio according to those universe-level sector weights. Within each represented sector, stocks are weighted by their own market capitalisation.

This structure is intended to preserve the broad sector composition of the eligible universe while reducing unintended concentration in a small number of dominant large-cap stocks. This is particularly relevant in the Vietnamese equity market, where the investment universe contains substantial heterogeneity in firm size, liquidity, sector composition, and trading activity.

Let $\mathcal{E}_t$ denote the eligible investment universe at formation date t, and let $\mathcal{P}_{k,t}$ denote the set of stocks assigned to portfolio k at date t, where $k \in \{P1, P2, P3\}$. Let $M_{i,t}$ denote the market capitalisation of stock i at formation date t, and let $s(i)$ denote the sector classification of stock i. The universe-level weight of sector s is first computed as:

$$W_{s,t}^{U} = \frac{\sum_{i \in \mathcal{E}_t,\, s(i)=s} M_{i,t}}{\sum_{i \in \mathcal{E}_t} M_{i,t}} \tag{6}$$

where $W_{s,t}^{U}$ is the market-capitalisation weight of sector s in the full eligible universe at formation date t.

For each target portfolio, the universe sector weights are then restricted to the sectors that are represented in that portfolio. Let $\mathcal{R}_t(k)$denote the set of sectors represented in portfolio k at date t. The target sector weight for sector s in portfolio k is:

$$W_{s,k,t}^{T} = \frac{W_{s,t}^{U}}{\sum_{s' \in \mathcal{R}_t(k)} W_{s',t}^{U}}, \quad s \in \mathcal{R}_t(k) \tag{7}$$

where s' is a dummy index that runs over all sectors represented in portfolio k. This renormalisation ensures that the represented sector weights sum to one within each portfolio-date pair.

Within each represented sector, stocks assigned to the target portfolio are then weighted according to their market capitalisation relative to the total market capitalisation of all stocks in the same portfolio-sector group:

$$w_{i,k,t} = W_{s(i),k,t}^{T} \times \frac{M_{i,t}}{\sum_{j \in \mathcal{P}_{k,t},\, s(j)=s(i)} M_{j,t}}, \quad i \in \mathcal{P}_{k,t} \tag{8}$$

where $w_{i,k,t}$ is the final portfolio weight of stock i in portfolio k at formation date t. The first component, $W_{s(i),k,t}^{T}$, determines the allocation to the stock's sector, while the second component determines the stock's weight within that sector based on market capitalisation.

The resulting stock weights are normalised to sum to one for each portfolio-date pair:

$$\sum_{i \in \mathcal{P}_{k,t}} w_{i,k,t} = 1 \tag{9}$$

For the value-only and momentum-only market-capitalisation-weighted modules, this weighting procedure is applied separately to P1, P2, and P3. For the combined value-momentum market-capitalisation-weighted module, the same weighting procedure is applied to the P1 portfolio used in the reported performance summary, while the stock assignment output still records the P1, P2, and P3 classifications for each formation date.

The market capitalisation used in the weighting procedure is the most recently available observation on or before the relevant formation date. For the quarterly value model, this corresponds to the value portfolio construction date after applying the 45-day publication lag to the fundamental data. For the weekly momentum and combined models, this corresponds to the weekly formation date based on the Thursday closing price signal, with rebalancing assumed to occur on the following Friday. This timing convention avoids the use of future information in portfolio construction.

5. Trading Cost, Turnover, and Return Computation

5.1 Trading Cost Model

The final implementation applies explicit transaction costs in the market-capitalisation-weighted value, momentum, and combined modules. The configured trading cost assumptions are a 0.15% buy-side brokerage fee, a 0.15% sell-side brokerage fee, and a 0.10% sell-side tax. The implied round-trip cost is therefore 0.40%.

Cost component	**Rate**
Buy brokerage fee	0.15%
Sell brokerage fee	0.15%
Sell tax	0.10%
Total round-trip cost	0.40%

At each rebalance, the model distinguishes entering stocks, exiting stocks, and continuing holdings. Entering stocks are bought at their current target weights, and the buy fee is applied to those weights. Exiting stocks are sold at their drifted pre-rebalance weights, meaning the previous portfolio weights are first grown by the holding-period returns and then renormalised before applying the sell fee and sell tax. Continuing holdings incur cost only on the active rebalance trade: an increase in target weight receives the buy fee, while a decrease in target weight receives the sell fee plus sell tax.

$$\mathrm{FeeDrag_{k,t}} = \mathrm{BuyCost_{k,t}} + \mathrm{SellCost_{k,t}} + \mathrm{RebalanceCost_{k,t}} \tag{10}$$

$$\mathrm{BuyCost_{k,t}} = \sum_{i \in B_{k,t}} w_{i,k,t} \times c^{B} \tag{11}$$

$$\mathrm{SellCost_{k,t}} = \sum_{i \in S_{k,t}} w'_{i,k,t} \times (c^{S} + \tau^{S}) \tag{12}$$

$$\text{RebalanceCost}_{\text{k,t}} = \sum_{i \in H_{k,t}} \left[max\left(w_{i,k,t} - w'_{i,k,t}, 0\right) c^B + max\left(w'_{i,k,t} - w_{i,k,t}, 0\right) \left(c^S + \tau^S\right) \right] \quad (13)$$

where $B_{k,t}$ is the set of stocks entering portfolio k at formation date t, $S_{k,t}$ is the set of stocks exiting the portfolio, and $H_{k,t}$ is the set of stocks remaining in the portfolio. $w_{i,k,t}$ denotes the current target weight of stock i, while $w'_{i,k,t}$ denotes the drifted pre-rebalance weight based on the previous portfolio weights and realised returns since the prior rebalance. c^B is the buy-side brokerage fee, c^S is the sell-side brokerage fee, and τ^S is the sell-side transaction tax.

For the first rebalance of a portfolio, the model assumes the full portfolio is bought from cash, so turnover is set to 100% and the buy fee is applied to the full target portfolio weight.

5.2 Turnover

Turnover is computed as a weight-based measure rather than a stock-count measure in the market-capitalisation-weighted implementation. At each rebalance, the entering weight is the sum of target weights for newly added stocks, the exiting weight is the sum of drifted pre-rebalance weights for removed stocks, and the holding adjustment is the sum of absolute differences between target weights and drifted weights for stocks that remain in the portfolio.

$$\text{Turnover}_{\text{k,t}} = \frac{1}{2}\left(\sum_{i \in B_{k,t}} w_{i,k,t} + \sum_{i \in S_{k,t}} w'_{i,k,t} + \sum_{i \in H_{k,t}} \left| w_{i,k,t} - w'_{i,k,t} \right| \right) \quad (14)$$

The division by two prevents double-counting the buy and sell sides of the same reallocation. This measure better reflects actual traded portfolio capital than a simple count of changed tickers.

5.3 Gross and Net Return Computation

Gross return is computed as the market-capitalisation-weighted average of constituent stock returns over the relevant holding period. Stocks with missing holding-period returns are excluded from the return calculation, and the remaining weights are effectively renormalised by dividing by the sum of weights with valid returns.

$$R_{k,t}^{gross} = \frac{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t} {}^{*} r_{i,t}}{\sum_{i \in \mathcal{V}_{k,t}} w_{i,k,t}} \tag{15}$$

$$R_{k,t}^{net} = R_{k,t}^{gross} - \mathrm{FeeDrag}_{\mathrm{k,t}} \tag{16}$$

where $R_{k,t}^{gross}$ is the gross return of portfolio k over holding period t, $R_{k,t}^{net}$ is the corresponding return after transaction costs, $w_{i,k,t}$ is the target portfolio weight of stock i, $r_{i,t}$ is the realised simple return of stock i over the holding period, and $\mathcal{V}_{k,t}$ denotes the subset of portfolio stocks with valid return observations. The denominator renormalises the portfolio weights when some constituent returns are missing.

For the value-only model, the holding-period return is computed directly from one quarterly safe-date rebalance to the next using the latest available daily prices on or before those dates. For the momentum-only and combined models, the holding-period return is computed from one Thursday weekly price to the next Thursday weekly price. Annualised performance in the output is calculated arithmetically: average quarterly return multiplied by four for the value model, and average weekly return multiplied by 52 for the weekly momentum and combined models. This is an arithmetic annualisation rather than a compounded annual growth rate.

6. Implementation Assumptions and Limitations

First, the framework is long-only. It does not implement short portfolios or long-short spread portfolios, even though the academic factor literature often evaluates value and momentum using long-short returns.

Second, the weekly momentum signal uses a one-week Thursday-to-Thursday return and assumes Friday rebalancing after the current Thursday close.

Finally, transaction costs are represented by fixed brokerage and tax rates. The model does not estimate bid-ask spreads, market impact, liquidity constraints, or execution slippage. The fee model is therefore a simplified implementation-cost approximation, consistent with the practical focus on turnover and fee drag emphasised by Frazzini, Israel and Moskowitz (2012).

7. References

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